

 POLITÉCNICO DE LEIRIA	Organisation: IPLeiria	Date: 08/02/2024
	Customer: NA	Project: Master Thesis
	Team: NA	Ref: MEAU-DataAcqBoard

Requirements Report

1 Job data

Type	Location	Reference
Requirements report	Leiria	MEAU-DataAcqBoard

Done by	Start date	End date
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Brief
The current document provides the established requirements for MEAU-DataAcqBoard project.

2 Requirements
2.1 Develop microcontroller based module (familiarity, ease of deployment, enough processing power)
2.2 Capable of acquiring at least 10 analog signals (NTCs) and 1 digital signal (flow meter)
2.3 Analog signal voltage reading error must not exceed 5%
2.4 Capable of storing the acquired data in permanent storage at 2Hz
2.5 Capable of displaying acquired data to user in a timely manner (test safety control in real time)
2.6 Hardware to enable manual control of data acquisition start/stop

3 Optional Features
3.1 Implementation of CAN Bus to allow future use as datalogger of data generated elsewhere
Power outputs (+12V) to allow board to drive cooling accessories. Water Pump output is to be used to generate a PWM signal to set pump speed only. (Output current rating: 1A) 3.2 Fan output is to be used to supply power to a fan with a current rating of 0.3A. (Output current rating: 1A) Extra power output may be used to power other loads. (Output current rating: 2A)